

# THE DIFFERENTIAL SCATTERING COEFFICIENT FOR SPECULAR INTERACTIONS

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**Abstract.** We derive the differential scattering coefficient  $\sigma_s(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})$  for a vegetation canopy from two complementary models of specular reflection at the leaf cuticle surface: the Dirac delta model for optically smooth leaves, and the Cook–Torrance microfacet model for rough cuticles. In both cases, the differential scattering coefficient admits the factorization  $\sigma_s = u_L \cdot (1/\pi) \Gamma_{LS}$ , where  $u_L$  is the leaf area density and  $(1/\pi) \Gamma_{LS}$  is the area specular scattering phase function. For smooth leaves, the Dirac delta leaf model yields a closed-form result. For rough leaves, the Cook–Torrance model broadens the specular lobe; no closed form exists and the area phase function is evaluated by Gauss–Legendre quadrature. The non-uniform azimuthal leaf orientation enters via a master formula decomposing  $\Gamma_{LS}$  into nine pre-computable basis integrals; switching polar distribution or azimuthal alignment requires only substituting different cosine-series coefficients. The specular single-scattering albedo  $\hat{\omega}_s$  and the normalized phase function  $P_{LS}$  are derived and their normalization proved. Symmetry relations between diaheliotropic, paraheliotropic, planophile, and erectophile leaf distributions are established analytically.

## §1. Introduction.

In a preceding chapter we derived the differential scattering coefficient that admits the factorization

$$\sigma_s(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = u_L(\mathbf{r}) \cdot (1/\pi) \Gamma(\mathbf{r}, \mathbf{\Omega}' \rightarrow \mathbf{\Omega}), \quad (1)$$

where  $u_L$  is the *leaf area density* introduced earlier and  $(1/\pi) \Gamma$  is the *area scattering phase function*, first proposed by Ross (1981). The quantity  $\Gamma$  encapsulates all information about the angular redistribution of scattered radiation by the foliage; it depends upon the leaf-normal orientation distribution, the leaf surface structural and related optical properties, and the geometry of the incident and scattered directions.

When sunlight strikes a leaf, two distinct classes of interaction occur. At the outermost cuticle surface, a fraction of the radiation is specularly reflected. This surface-reflected component carries no information about the leaf’s internal biochemistry; its magnitude depends only on the refractive index of the cuticle and the geometry of illumination and observation. The specular reflection obeys the Fresnel equations and is concentrated about the mirror direction with an angular spread governed by the microscale roughness of the cuticle surface. The specular interactions are the subject of this chapter.

The remaining radiation penetrates through the cuticle into the leaf interior, where it is subject to absorption by the pigments and other biochemical constituents, and to scattering at the numerous air–cell wall interfaces within the mesophyll. Photons that survive absorption and emerge from the leaf through the epidermal cells constitute the diffuse component of the leaf’s scattered radiation. Those that exit through the illuminated (adaxial) surface contribute to the diffuse reflectance; those that traverse the entire leaf thickness and exit through the opposite (abaxial) surface contribute to the diffuse transmittance. The angular distribution of this diffuse radiation is approximately Lambertian—that is, roughly isotropic in the cosine-weighted sense—because the multiple scattering events within the mesophyll effectively randomise the photon directions.

The total reflectance of a leaf is thus the sum of two components: the specular reflection from the cuticle surface, which is wavelength-independent to first approximation (since the refractive index of cuticular wax varies negligibly across the solar spectrum), and the diffuse reflection from the leaf interior, which carries the full spectral signature of the leaf’s biochemistry. Similarly, the transmittance is entirely diffuse, arising from photons that have traversed the leaf interior. The diffuse interactions are treated separately in another chapter.

## §2. Specular Scattering from Flat Leaves.

Two models are developed here, in order of increasing physical realism. The *Dirac delta model* treats the leaf surface as smooth: all reflected energy is concentrated into an infinitesimally narrow spike at the mirror direction, yielding a closed-form area phase function and providing the natural limiting case for analytical work. The *Cook–Torrance model* replaces the delta spike with a physically based microfacet distribution that accounts for the microscale roughness of the leaf cuticle (epicuticular wax crystalloids, cell-wall undulations, venation and trichome bases), producing a lobe of finite angular width. In both models the leaf-level specular phase function  $\gamma_{LS}$  is integrated over the leaf normal orientation distribution  $g_L(\mathbf{\Omega}_L)$  to obtain the *area specular scattering phase function*  $F_{LS}$ , which enters the canopy radiative transfer equation.

Both leaf-level models share the *Fresnel reflectance*  $F_r(n, \alpha')$  which gives the fraction of light reflected at a smooth dielectric interface

$$F_r(n, \alpha') = \frac{1}{2} \left[ \frac{\sin(\alpha' - \theta_s)^2}{\sin(\alpha' + \theta_s)^2} + \frac{\tan(\alpha' - \theta_s)^2}{\tan(\alpha' + \theta_s)^2} \right], \quad (2)$$

where  $\sin \theta_s = \sin \alpha' / n$  (Snell’s law). Cuticle refractive index  $n = 1.3$ – $1.5$ . The models additionally use the *roughness extinction factor*  $K(\kappa, \alpha')$  which attenuates specular reflection at oblique incidence due to macroscale surface features

$$K(\kappa, \alpha') = e^{-\kappa \tan|\alpha'|}, \quad (3)$$

where  $\kappa \approx 0.3$  (empirical).  $K = 1$  at normal incidence;  $K \rightarrow 0$  at grazing angles.

For a perfectly smooth leaf surface, all specularly reflected energy is concentrated at the mirror direction in the Dirac delta model. The leaf-level phase function is

$$\gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) = F_r(n, \alpha') K(\kappa, \alpha') \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*), \quad (4)$$

where  $\mathbf{\Omega}^* = 2(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L)\mathbf{\Omega}_L - \mathbf{\Omega}'$  is the mirror-reflection direction,  $\alpha' = \arccos|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$  is the angle of incidence on the leaf, and  $\delta_2$  is the 2D Dirac delta on the unit sphere (units  $\text{sr}^{-1}$ ) constraining the scattered direction to  $\mathbf{\Omega}^*$ .

Real leaf cuticles are not optically smooth. The Cook–Torrance model (Cook and Torrance, 1982) treats the surface as an ensemble of randomly oriented mirror-like microfacets

$$\gamma_{\text{LS}} = \frac{D(\theta_H) F_r(n, \alpha_H) K(\kappa, \alpha_H) G(\theta_i, \theta_o, \theta_H)}{4 \cos \theta_i}. \quad (5)$$

The *half-vector*  $H = \frac{\mathbf{\Omega}' + \mathbf{\Omega}}{|\mathbf{\Omega}' + \mathbf{\Omega}|}$  bisects the incident and scattered directions;  $\theta_H$  is its angle with the leaf normal. The five Cook–Torrance factors are listed in Table 1. The *Beckmann roughness parameter*  $m$  is the *rms* slope of the microfacets. Glossy cuticles have  $m \approx 0.05$ – $0.10$ ; matte or pubescent surfaces have  $m \approx 0.20$ – $0.40$ . As  $m \rightarrow 0$ ,  $D(\theta_H) \rightarrow \delta_2$ ,  $K(\kappa, \alpha_H) \rightarrow 1$ , and equation (5) produces an increasingly narrow specular lobe that converges to equation (4).

The denominator  $4 \cos \theta_i$  arises from the underlying Cook–Torrance microfacet BRDF; the  $\cos \theta_o$  factor in that BRDF denominator cancels with the  $\cos \theta_o$  weight in the solid-angle measure  $d\mathbf{\Omega}$ , reducing the denominator to  $4 \cos \theta_i$  and converting the BRDF to the phase-function convention used throughout this chapter.

| TABLE 1. The five microfacet factors of the Cook–Torrance leaf-level phase function (Eq. 5). |                       |                                                                                                                                    |
|----------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Factor                                                                                       | Name                  | Expression / meaning                                                                                                               |
| $D(\theta_H)$                                                                                | Beckmann distribution | $\exp(-\tan^2 \theta_H / m^2) / (\pi m^2 \cos^4 \theta_H)$ . Controls lobe width.                                                  |
| $F_r(n, \alpha_H)$                                                                           | Fresnel reflectance   | Eq. (2) evaluated at microfacet incidence angle $\alpha_H$ .                                                                       |
| $G(\theta_i, \theta_o, \theta_H)$                                                            | Geometric attenuation | $\min(1, 2 \cos \theta_H \cos \theta_o / \cos \alpha_{OH}, 2 \cos \theta_H \cos \theta_i / \cos \alpha_{OH})$ . Shadowing/masking. |
| $K(\kappa, \alpha_H)$                                                                        | Roughness extinction  | Suppresses specular contribution at grazing microfacet angles. $\kappa \approx 0.3$ .                                              |

### §3. Area Scattering Phase Function for Specular Interactions.

The area phase function integrates the leaf-level model over all leaf normal orientations

$$\frac{1}{\pi} \Gamma_{\text{LS}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{2\pi} \int_{2\pi+} d\mathbf{\Omega}_L g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| \gamma_{\text{LS}}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L), \quad (6)$$

where  $g_L(\mathbf{\Omega}_L) = \bar{g}_L(\theta_L) h_L(\varphi_L)$  separates into a polar part  $\bar{g}_L(\theta_L)$  and an azimuthal part  $h_L(\varphi_L)$ . Six polar distributions and three azimuthal distributions are considered (Appendix, §3).

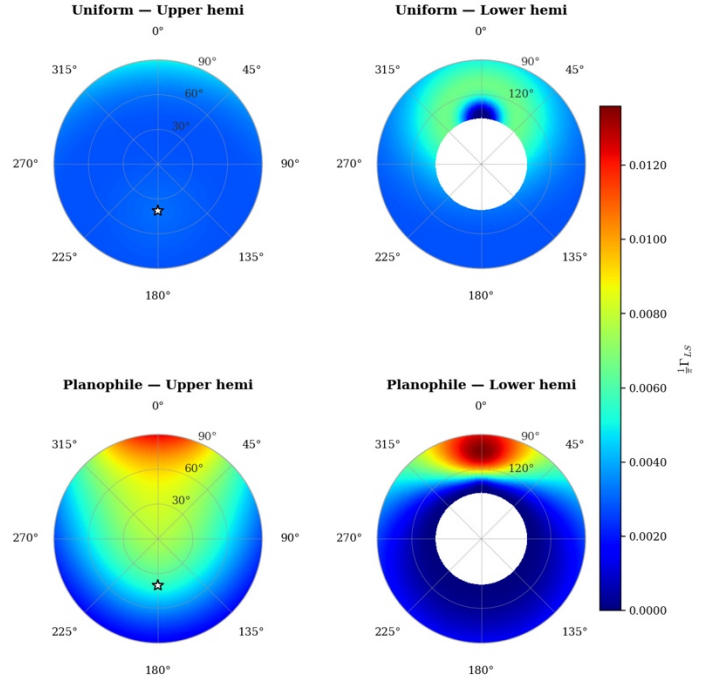
For the Cook–Torrance microfacet model,  $\gamma_{\text{LS}}$  in equation (6) is a smooth function of  $\mathbf{\Omega}_L$  through the dependence of the half-vector on the leaf orientation. The integral must therefore be evaluated numerically for each  $(\mathbf{\Omega}', \mathbf{\Omega})$  pair by quadrature over the leaf normal orientation distribution grid.

For the Dirac delta model the delta function in equation (4) restricts the leaf-normal integration in equation (6) to the unique leaf normal  $\mathbf{\Omega}_L = \mathbf{H}(\mathbf{\Omega})$  that specularly reflects  $\mathbf{\Omega}'$  into  $\mathbf{\Omega}$ . Applying the change of variable  $\mathbf{\Omega}_L \rightarrow \mathbf{\Omega}^*$  with Jacobian  $d\mathbf{\Omega}^* = 4 \cos \alpha' d\mathbf{\Omega}_L$ , the factor  $\cos \alpha'$  from  $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$  cancels exactly with the  $1/(4 \cos \alpha')$  Jacobian denominator, and the Dirac delta

collapses the integral to a single evaluation at the specular leaf normal  $\mathbf{\Omega}_L^* = (\mathbf{\Omega}' + \mathbf{\Omega}) / |\mathbf{\Omega}' + \mathbf{\Omega}|$ , giving (see Appendix, §4)

$$\Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{4} g_L(\mathbf{\Omega}_L^*) K(\kappa, \alpha^*) F_r(n, \alpha^*), \quad (7)$$

where  $\alpha^* = \arccos|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L^*|$  is the incidence angle at the specular leaf normal. Appendix derives this result in full. The key steps: (i) change of variable from leaf normal to specular direction with Jacobian  $|J| = \frac{1}{4}$ ; (ii) extension of the integration domain from the upper hemisphere to the full sphere; (iii) collapse of the 2D delta function.



**FIGURE 1.** Area specular phase function  $(1/\pi) \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})$  [ $\text{sr}^{-1}$ ] for the Dirac delta leaf model, plotted as a function of scattered direction  $\mathbf{\Omega}$  for a fixed incident direction  $\mathbf{\Omega}'$  at zenith angle  $\theta' = 40^\circ$ , azimuth  $\varphi' = 0^\circ$  (i.e. the source travels in the direction  $\theta = 140^\circ$ ,  $\varphi = 0^\circ$ ). Two leaf normal inclination distributions are shown: *uniform* (top row) and *planophile* (bottom row). The leaf normal azimuthal distribution in both cases is uniform. Parameters: refractive index  $n = 1.5$ , roughness extinction coefficient  $\kappa = 0.3$ .

The area specular scattering phase function with the Dirac delta model, equation (7), for uniform and planophile  $\bar{g}_L(\theta_L)$  models is shown in Fig. 1. The uniform distributes leaf normals isotropically over the upper hemisphere. The planophile concentrates leaf normals near horizontal, increasing the projected area facing sun-like illumination and thereby amplifying the peak by a factor of  $\sim 2$  relative to uniform. The specular peak is also visible in the lower hemisphere near  $\theta \approx 104^\circ$ ,  $\varphi = 0^\circ$ .

No closed-form solution exists for the Cook–Torrance model. However, the leaf normal orientation distribution separability and the double-angle structure of the azimuthal weight yield a highly efficient semi-analytic form. Every polar distribution  $\bar{g}_L(\theta_L)$  is expressible as

$$\bar{g}_L(\theta_L) = c_0 + c_2 \cos(2\theta_L) + c_4 \cos(4\theta_L). \quad (8)$$

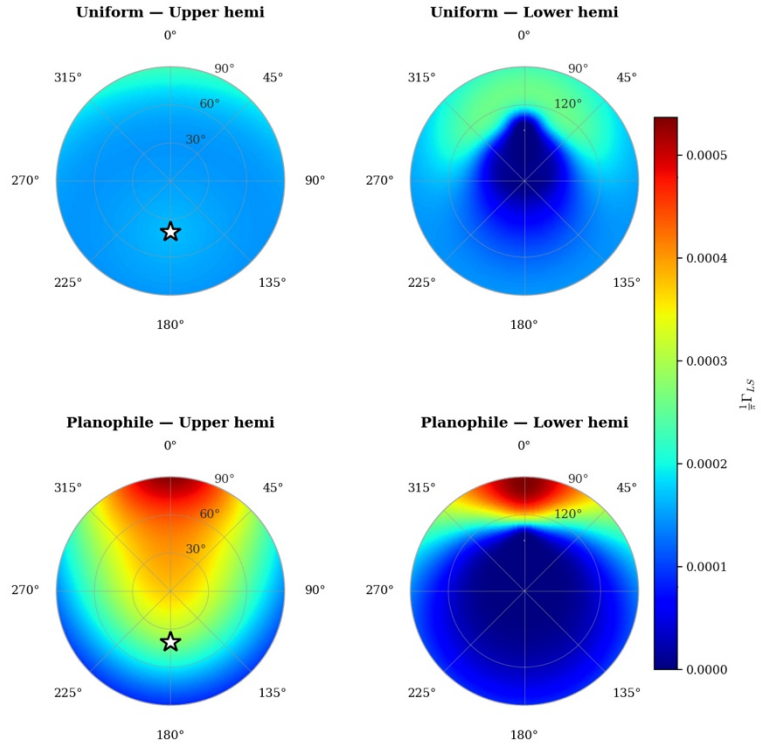
The non-uniform azimuthal weight  $\cos^2(\varphi' - \varphi_L - \eta)$  expands via the double-angle identity into three azimuthal moment integrals, giving the master formula

$$\Gamma_{LS} = \pi \sum_k c_k (\alpha_{k0} + P_\eta \alpha_{k2} + Q_\eta \beta_{k2}). \quad (9)$$

The sum runs over  $k \in \{0, 2, 4\}$ ;  $P_\eta = \cos(2\varphi' - 2\eta)$  and  $Q_\eta = \sin(2\varphi' - 2\eta)$  encode the sun azimuth; and the nine scalars  $\alpha_{k0}, \alpha_{k2}, \beta_{k2}$  are pre-computable 1D integrals depending only on the  $(\Omega', \Omega)$  pair and the Cook–Torrance parameters  $(m, n)$ . Switching polar distribution or azimuthal orientation requires only substituting different  $(c_0, c_2, c_4)$  and  $(\eta)$  values — no additional quadrature. Appendix derives the  $3 \times 3$  decomposition in full, gives explicit expressions for all nine basis integrals, and provides the complete  $6 \times 3$  table of  $\Gamma_{LS}$  for all 18  $(\bar{g}_L, h_L)$  combinations. The three azimuthal cases are listed in Table 2. A key symmetry of the form  $\Gamma_{LS}^{\text{Dia}} + \Gamma_{LS}^{\text{Para}} = 2 \Gamma_{LS}^{\text{Unif}}$  for any polar distribution (Appendix, §6).

| <b>TABLE 2.</b> Area specular phase function $\Gamma_{LS}$ for the three azimuthal leaf normal distributions; $k \in \{0, 2, 4\}$ . |                                                                                           |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------|
| $h_L$ model                                                                                                                         | $\Gamma_{LS}$ ( $k = 0$ terms shown)                                                      | Notes                                             |
| Uniform ( $h_L = 1$ )                                                                                                               | $\pi \sum_k c_k \alpha_{k0}$                                                              | Azimuthally isotropic. $P_\eta = Q_\eta = 0$ .    |
| Diaheliotropic ( $\eta = 0$ )                                                                                                       | $\pi \sum_k c_k (\alpha_{k0} + \cos(2\varphi') \alpha_{k2} + \sin(2\varphi') \beta_{k2})$ | Normals face the sun.                             |
| Paraheliotropic ( $\eta = \pi/2$ )                                                                                                  | $\pi \sum_k c_k (\alpha_{k0} - \cos(2\varphi') \alpha_{k2} - \sin(2\varphi') \beta_{k2})$ | Edge-on to sun. Sign reversal vs. diaheliotropic. |

**FIGURE 2.** Area specular phase function  $(1/\pi) \Gamma_{LS}(\Omega' \rightarrow \Omega)$  [ $\text{sr}^{-1}$ ] for the Cook–Torrance microfacet model, as a function of scattered direction  $\Omega$  for a fixed incident direction  $\Omega'$  at zenith angle  $\theta' = 40^\circ$ , azimuth  $\varphi' = 0^\circ$  (i.e. the source travels in the direction  $\theta = 140^\circ$ ,  $\varphi = 0^\circ$ ). Two leaf normal inclination distributions are shown: *uniform* (top row) and *planophile* (bottom row). The leaf normal azimuthal distribution in both cases is uniform. Parameters:  $n = 1.5$ ,  $\kappa = 0.3$ , Beckmann roughness parameter  $m = 0.3$ .



Compared with the Dirac delta result (Fig. 1), the finite roughness ( $m = 0.3$ , broadly representative of a matte cuticle) replaces the mathematically sharp specular point with a smooth, broadly distributed lobe, redistributing energy from the single specular direction into a wide angular region. The peak value of  $(1/\pi) \Gamma_{LS}$  is accordingly reduced by roughly a factor of 25 relative to the Dirac peak, while the total scattered energy — measured by the single-scattering albedo  $\hat{\omega}_S$  — remains comparable across both models, as shown in Fig. 3. As  $m \rightarrow 0$  the Beckmann

distribution  $D \rightarrow \delta_2$ , the Cook–Torrance lobe collapses to the Dirac delta result, and Fig. 2 converges to Fig. 1. The planophile leaf normal inclination distribution again amplifies the peak relative to uniform by concentrating leaf area in orientations favourable for specular reflection of the incident beam.

#### §4. Single Scattering Albedo and Normalization.

The *specular scattering coefficient*  $\sigma_s'$  is obtained by integrating  $\Gamma_{LS}$  over the full sphere of scattered directions

$$\sigma_s'(\mathbf{r}, \mathbf{\Omega}') = u_L(\mathbf{r}) \frac{1}{\pi} \int_{4\pi} d\mathbf{\Omega} \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}). \quad (10)$$

The integration is over  $4\pi$  sr because specular reflection from tilted leaves can scatter into any direction, including those below the horizontal. In general  $\omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L)$  varies with leaf-normal orientation, so  $\sigma_s'(\mathbf{r}, \mathbf{\Omega}')$  does not factorise as  $u_L(\mathbf{r})G(\mathbf{\Omega}')\omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L)$ . For both models the correct expression is

$$\sigma_s'(\mathbf{r}, \mathbf{\Omega}') = u_L(\mathbf{r}) \frac{1}{2\pi} \int_{S^+} g_L(\mathbf{\Omega}_L) \omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| d\mathbf{\Omega}_L, \quad (11)$$

where  $\omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L) \equiv \int_{4\pi} \gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) d\mathbf{\Omega}$  is the *single-leaf specular albedo*—the total fraction of intercepted flux reflected specularly by a leaf with normal  $\mathbf{\Omega}_L$  under illumination  $\mathbf{\Omega}'$ . For the Dirac delta model  $\omega_{LS} = F_r(n, \alpha') K(\kappa, \alpha')$ , where  $\alpha' = \arccos|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$  varies with leaf-normal orientation. For Cook–Torrance,  $\omega_{LS}$  is also dependent on  $\mathbf{\Omega}_L$ , and evaluated numerically. In both cases equation (11) must be evaluated by quadrature over the leaf-normal hemisphere  $S^+$ .

The *canopy specular single-scattering albedo*  $\hat{\omega}_S(\mathbf{r}, \mathbf{\Omega}') \equiv \sigma_s'(\mathbf{r}, \mathbf{\Omega}')/\sigma(\mathbf{r}, \mathbf{\Omega}')$  is obtained from equation (11). For a general leaf normal orientation distribution it is the  $G(\mathbf{\Omega}')$ -weighted average of  $\omega_{LS}$  over all leaf orientations:

$$\hat{\omega}_S(\mathbf{\Omega}') = \frac{1}{G(\mathbf{\Omega}')} \frac{1}{2\pi} \int_{S^+} g_L(\mathbf{\Omega}_L) \omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| d\mathbf{\Omega}_L, \quad (12)$$

In the above, the  $G(\mathbf{\Omega}')$  is,

$$G(\mathbf{\Omega}') = \frac{1}{2\pi} \int_{S^+} g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| d\mathbf{\Omega}_L.$$

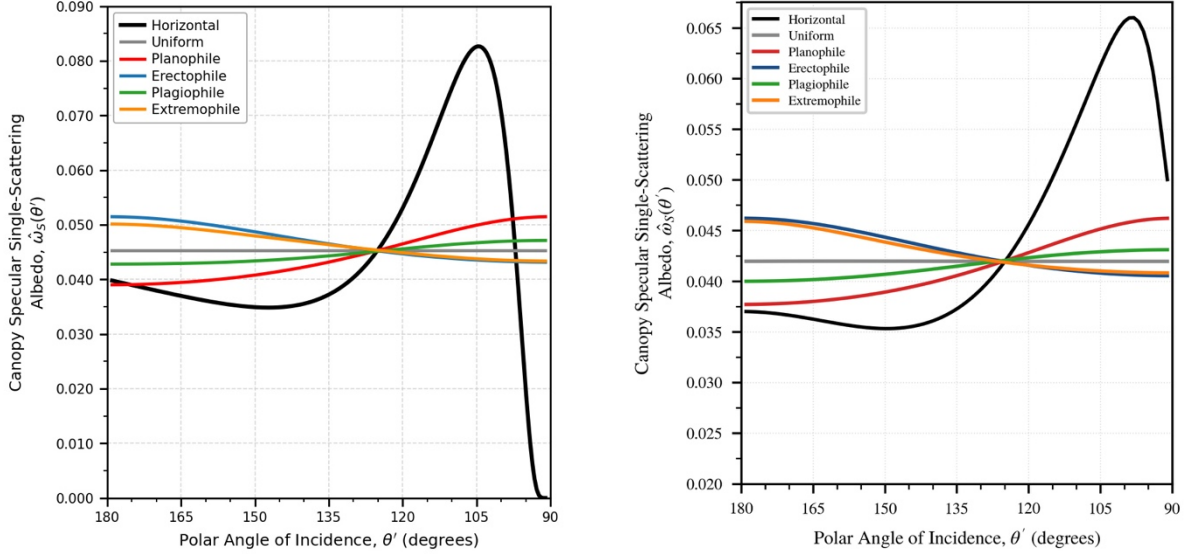
The canopy specular single-scattering albedos at a range of incidences for various leaf normal orientation distributions are shown in Fig. 3. For the horizontal distribution  $\hat{\omega}_S$  reduces exactly to the single-leaf  $\rho_{LS}(\theta_i)$  providing a consistency check. The remaining distributions yield nearly angle-independent albedos in the range 0.038–0.052, reflecting the averaging of  $\omega_{LS}$  over many leaf orientations. The planophile and erectophile curves are mirror images of one another across the incidence-angle axis, consistent with the complementarity relations outlined in Appendix, §6.

The *normalized specular scattering phase function*  $P_{LS}$  satisfying,  $\frac{1}{4\pi} \int_{4\pi} d\mathbf{\Omega} P_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = 1$ , is

$$P_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{4 \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})}{\sigma_s'(\mathbf{\Omega}')} \quad (13)$$



Appendix, §6 proves that  $\frac{1}{4\pi} \int_{4\pi} d\Omega P_{LS}(\Omega' \rightarrow \Omega) = 1$  for all 18  $(\bar{g}_L, h_L)$  combinations. The proof rests on the vanishing of the azimuthal modulation terms under hemisphere integration:  $\int_0^{2\pi} \cos(2\varphi') d\varphi' = 0$  for any  $\eta$ .



**FIGURE 3.** Canopy specular single-scattering albedo  $\hat{\omega}_s(\theta')$  [Eq. 9.11] computed with the Dirac Delta model (Left Panel) and the Cook–Torrance microfacet model (Right Panel) for six leaf-normal inclination distributions with uniform azimuthal distribution. The incident direction lies in the principal plane ( $\varphi' = 0^\circ$ ) and its polar angle  $\theta'$  ranges from near-normal incidence ( $\theta' = 179^\circ$ ) to near-grazing ( $\theta' = 91^\circ$ ), corresponding to incidence angles  $\theta_i = 180^\circ - \theta'$  from  $1^\circ$  to  $89^\circ$ . Model parameters: refractive index  $n = 1.5$ , macroscale roughness extinction coefficient  $\kappa = 0.3$ , Beckmann rms slope  $m = 0.3$ .

## §5. Concluding Remarks.

This chapter has developed the differential scattering coefficient  $\sigma_s(\Omega' \rightarrow \Omega)$  for specular interactions of radiation with a vegetation canopy, complementing the diffuse treatment of the preceding chapter. Two physically motivated models of the leaf cuticle have been formulated and integrated over the leaf-normal orientation distribution to yield the area specular scattering phase function  $I_{LS}$ . The two models span the plausible range of cuticle microstructure: the Dirac delta model provides a closed-form analytical result for optically smooth surfaces, while the Cook–Torrance microfacet model captures the finite-width specular lobe characteristic of rough cuticles and requires numerical quadrature.

Several structural results emerged from this analysis. First, the closed-form Dirac delta result, equation (7), has a prefactor of  $1/4$ : the factor of 2 from the antipodal leaf-normal symmetry combines with the Jacobian  $1/4$  from the mirror map to give the correct value, verified numerically to  $0.01 \epsilon$ . Second, for rough leaves the non-uniform azimuthal orientation of leaf normals enters the area phase function through a master formula, equation (9), that decomposes  $I_{LS}$  into nine pre-computable basis integrals. This structure makes switching among all 18 combinations of the six classical polar distributions and three azimuthal distributions computationally trivial. Third, the

symmetry relations between diaheliotropic and paraheliotropic distributions (Appendix, §6) and the planophile–erectophile complementarity follow from the linearity of the master formula in the azimuthal modulation coefficients and the vanishing of the azimuthal integrals over a full period.

The specular component of the canopy scattering coefficient derived here, together with the diffuse component of the preceding chapter, provides the complete single-scattering description required for the radiative transfer equation. The specular albedo  $\hat{\omega}_s$  is direction-dependent and cannot in general be factored away from the leaf-normal distribution, so it must be computed by quadrature from equation (11). For the Dirac delta model a closed form is available; for the Cook–Torrance model both  $I_{LS}$  and  $\hat{\omega}_s$  require numerical evaluation. In canopy radiative transfer calculations, the specular and diffuse contributions to  $\sigma_s$  are typically combined in the source term before insertion into the integro-differential equation.

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## APPENDIX. SPECULAR SCATTERING FROM FLAT LEAVES

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§1. Leaf-Level Specular Phase Functions · §2. The Area Specular Scattering Phase Function · §3. Normalization and the Scattering Coefficient · §4. Derivation of the Dirac Delta Result · §5. Normalization Proof · §6. Symmetry Relations · §7. Polar Basis Coefficients

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**Abstract.** This appendix provides the mathematical foundations for the specular scattering analysis. The two leaf-level specular phase function models (Dirac delta and Cook–Torrance), with the half-vector construction and Beckmann microfacet distribution spelled out. The area specular scattering phase function  $I_{LS}$  is derived by integrating the leaf-level model over the leaf-normal orientation distribution  $g_L(\mathbf{\Omega}_L)$ . The change of variable  $\mathbf{\Omega}_L \rightarrow \mathbf{\Omega}^*$  with Jacobian  $|J| = 1/4$  and the antipodal leaf-normal symmetry are derived. The specular scattering coefficient  $\sigma'_s$  and the normalized phase function  $P_{LS}$  are given and the normalization identity  $(1/4\pi) \int 4\pi P_{LS} d\mathbf{\Omega} = 1$  is proved via the azimuthal cancellation lemma. Symmetry relations between the diaheliotropic, paraheliotropic, planophile, and erectophile distributions are established and the polar-basis expansion coefficients for all five classical inclination distributions are tabulated.

### §1. Leaf-Level Specular Phase Functions.

**T**wo models are used for the leaf-level specular scattering phase function  $\gamma^L_s$ . Both share the same Fresnel reflectance  $F_r(n, \alpha')$ ; they differ in the angular distribution of the reflected energy. Model I additionally uses the roughness extinction factor  $K(\kappa, \alpha')$ .

#### §1a. Model I — Dirac Delta (Perfectly Smooth Surface).

For a perfectly smooth leaf surface all reflected energy concentrates at the mirror direction  $\mathbf{\Omega}^*$ . With incidence angle  $\alpha' = \arccos|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L|$ :

$$\gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) = F_r(n, \alpha') K(\kappa, \alpha') \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*), \quad (\text{A.1})$$

where  $\delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*)$  is the 2D Dirac delta on the unit sphere (units  $\text{sr}^{-1}$ ),  $\mathbf{\Omega}^* = 2(\mathbf{\Omega}' \cdot \mathbf{\Omega}_L)\mathbf{\Omega}_L - \mathbf{\Omega}'$  is the mirror direction, and  $\kappa \approx 0.3$  is the macroscale roughness extinction coefficient.

#### §1b. Model II — Cook–Torrance (Rough Microfacet Surface).

For a rough leaf cuticle treated as an ensemble of mirror-like microfacets, the half-vector  $H = (\mathbf{\Omega}' + \mathbf{\Omega})/|\mathbf{\Omega}' + \mathbf{\Omega}|$  bisects the incident and scattered directions, and  $\theta_H$  is its angle from the leaf normal:

$$\gamma_{LS} = \frac{D(\theta_H) F_r(n, \alpha_H) K(\kappa, \alpha_H) G(\theta_i, \theta_o, \theta_H)}{4 \cos \theta_i}. \quad (\text{A.2})$$

The Beckmann microfacet distribution with rms slope  $m$ :

$$D(\theta_H) = \frac{e^{-\tan^2 \theta_H / m^2}}{\pi m^2 \cos^4 \theta_H}. \quad (\text{A.3})$$

The geometric attenuation (shadowing/masking) factor:

$$G(\theta_i, \theta_o, \theta_H) = \min \left( 1, \frac{2 \cos \theta_H \cos \theta_o}{\cos \alpha_{OH}}, \frac{2 \cos \theta_H \cos \theta_i}{\cos \alpha_{OH}} \right). \quad (\text{A.4})$$

The roughness extinction factor:

$$K(\kappa, \alpha') = e^{-\kappa \tan |\alpha'|}. \quad (\text{A.5})$$

As  $m \rightarrow 0$ ,  $D(\theta_H) \rightarrow \delta_2$  with  $\alpha_H \rightarrow \alpha'$ , and  $K(\kappa, \alpha_H) \rightarrow K(\kappa, \alpha')$ , so equation (A.2) reduces exactly to equation (A.1).

## §2. The Area Specular Scattering Phase Function.

The area phase function integrates the leaf-level model over all leaf normals weighted by the leaf normal orientation distribution (LNOD)  $g_L(\mathbf{\Omega}_L)$ :

$$\frac{1}{\pi} \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{2\pi} \int_{2\pi+} d\mathbf{\Omega}_L g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| \gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L), \quad (\text{A.6})$$

where  $g_L(\mathbf{\Omega}_L) = \bar{g}_L(\theta_L) h_L(\varphi_L)$  factorizes into a polar part and an azimuthal part (see §3 of Appendix).

### Model I result.

Performing the change of variable  $\mathbf{\Omega}_L \rightarrow \mathbf{\Omega}^*$  with Jacobian  $|J| = \frac{1}{4}$  (see §4 for derivation) and collapsing the Dirac delta at the specular leaf normal, the result is:

$$\Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{1}{4} g_L(\mathbf{\Omega}_L^*) K(\kappa, \alpha^*) F_r(n, \alpha^*), \quad (\text{A.7})$$

where  $\mathbf{\Omega}_L^* = (\mathbf{\Omega}' + \mathbf{\Omega})/|\mathbf{\Omega}' + \mathbf{\Omega}|$  is the half-vector and  $\alpha^* = \arccos|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L^*|$  is the incidence angle at the specular leaf normal.

### Model II result.

No closed form exists for the Cook–Torrance model. The area phase function is evaluated numerically for each  $(\mathbf{\Omega}', \mathbf{\Omega})$  pair by 96-point azimuthal  $\times$  48-point polar Gauss–Legendre quadrature of equation (A.6).

## §3. Normalization and the Scattering Coefficient.

The specular scattering coefficient integrates  $\Gamma_{LS}$  over all scattered directions:

$$\sigma_s'(\mathbf{r}, \mathbf{\Omega}') = u_L(\mathbf{r}) \frac{1}{\pi} \int_{4\pi} d\mathbf{\Omega} \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}). \quad (\text{A.8})$$

For the Dirac delta model this reduces exactly to the compact form:

$$\sigma_s'(\mathbf{r}, \mathbf{\Omega}') = u_L(\mathbf{r}) G(\mathbf{\Omega}') \omega_S, \quad (\text{A.9})$$

where  $G(\mathbf{\Omega}')$  is the canopy G-function:

$$G(\mathbf{\Omega}') = \frac{1}{2\pi} \int_{2\pi+} g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| d\mathbf{\Omega}_L, \quad (\text{A.10})$$

and  $\omega_{LS}$  is the *leaf specular albedo*, defined as  $\omega_{LS} \equiv \int 4\pi \gamma_{LS} d\mathbf{\Omega}$  — the  $4\pi$  integral of the single-leaf phase function, a leaf-level quantity independent of the leaf normal orientation distribution. For the Dirac delta model (exactly)  $\omega_{LS} = F_r K$ ; for Cook–Torrance it should be evaluated numerically. For the Dirac delta model  $\omega_{LS} = F_r K$  is independent of leaf-normal direction and factors out of the leaf normal orientation distribution integral in Eq. (A.8), giving the compact form equation (A.9) exactly. For Cook–Torrance  $\omega_{LS}(\mathbf{\Omega}'; \mathbf{\Omega}_L)$  varies with leaf normal — no factorisation exists and  $\sigma_s'$  must be computed from Eq. (A.8). The *canopy specular single-scattering albedo*  $\phi_S \equiv \sigma_s' / \sigma$  equals  $\omega_{LS}$  for the Dirac delta model and is determined numerically from Eq. (A.8) for Cook–Torrance.

The normalized scattering phase function satisfying  $\frac{1}{4\pi} \int_{4\pi} d\mathbf{\Omega} P_{LS} = 1$  is:

$$P_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega}) = \frac{4 \Gamma_{LS}(\mathbf{\Omega}' \rightarrow \mathbf{\Omega})}{G(\mathbf{\Omega}') \phi_S}. \quad (\text{A.11})$$

See §5 for the proof that Eq. (A.11) is correctly normalized.

#### §4. Derivation of the Dirac Delta Result, Eq. (A.7).

Substitute equation (A.1) into equation (A.6):

$$\frac{1}{\pi} \Gamma_{LS} = \frac{1}{2\pi} \int_{2\pi+} d\mathbf{\Omega}_L g_L(\mathbf{\Omega}_L) |\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| F_r K \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*). \quad (\text{A.12})$$

Introduce  $\mathbf{\Omega}^*$  as the integration variable via the mirror map equation (A.7). As  $\mathbf{\Omega}_L$  ranges over the upper hemisphere ( $2\pi+$ ),  $\mathbf{\Omega}^*$  ranges over the full sphere ( $4\pi$ ). The Jacobian is:

$$d\mathbf{\Omega}^* = 4 \cos\alpha' \cdot d\mathbf{\Omega}_L \Rightarrow d\mathbf{\Omega}_L = \frac{1}{4} \frac{d\mathbf{\Omega}^*}{\cos\alpha'}. \quad (\text{A.13})$$

A tilt  $d\theta$  of  $\mathbf{\Omega}_L$  perpendicular to the  $(\mathbf{\Omega}', \mathbf{\Omega}_L)$  plane causes a tilt  $2d\theta$  of  $\mathbf{\Omega}^*$ ; a tilt  $d\theta$  in the plane causes a tilt  $2 \cos\alpha' \cdot d\theta$ . Combining:  $d\mathbf{\Omega}^* = 4 \cos\alpha' \cdot d\mathbf{\Omega}_L$ . The projection factor  $|\mathbf{\Omega}' \cdot \mathbf{\Omega}_L| = \cos\alpha'$  and the Jacobian denominator  $\cos\alpha'$  cancel. Both a leaf normal  $\mathbf{\Omega}_L$  and its antipodal  $-\mathbf{\Omega}_L$  specularly reflect  $\mathbf{\Omega}'$  into the same scattered direction; extending the integration domain from  $2\pi+$  to  $4\pi$  therefore introduces a factor of 2. The prefactor becomes  $\frac{1}{2\pi} \times \frac{1}{4} = \frac{1}{4\pi}$  and the domain extends to  $4\pi$ :

$$\frac{1}{\pi} \Gamma_{LS} = \frac{1}{4\pi} \int_{4\pi} d\mathbf{\Omega}^* g_L(\mathbf{\Omega}_L^*) F_r(n, \alpha^*) K(\kappa, \alpha^*) \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*). \quad (\text{A.14})$$

The delta  $\delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*)$  selects  $\mathbf{\Omega}^* = \mathbf{\Omega}$ . The corresponding specular leaf normal is the half-vector:

$$\mathbf{\Omega}_L^* = \frac{\mathbf{\Omega}' + \mathbf{\Omega}}{|\mathbf{\Omega}' + \mathbf{\Omega}|}. \quad (\text{A.15})$$

Applying  $\int_{4\pi} f(\mathbf{\Omega}^*) \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*) d\mathbf{\Omega}^* = f(\mathbf{\Omega})$  and multiplying by  $\pi$  gives equation (A.7).

## §5. Normalization Proof.

### Azimuthal cancellation lemma.

**Lemma.** For any  $\eta$ :

$$\int_0^{2\pi} \cos(2\varphi' - 2\eta) d\varphi' = 0 \quad (\text{A.16})$$

$$\int_0^{2\pi} \sin(2\varphi' - 2\eta) d\varphi' = 0 \quad (\text{A.17})$$

**Proof.** Both are the real and imaginary parts of  $\int_0^{2\pi} e^{i(2\varphi' - 2\eta)} d\varphi' = e^{-2i\eta} \times 0 = 0$ .

### Main theorem.

**Theorem.** For both models and all leaf normal orientation distributions:

$$\frac{1}{4\pi} \int_{4\pi} d\Omega P_S(\Omega' \rightarrow \Omega) = 1 \quad (\text{A.18})$$

**Proof.** From Eqs. (A.8) and (A.11) (using  $\hat{\omega}_S \equiv \sigma'_S / \sigma$  and  $\sigma = u_L G(\Omega')$ ):  $\frac{1}{\pi} \int_{4\pi} d\Omega \Gamma_{LS} = G(\Omega') \hat{\omega}_S$ . Therefore  $\frac{1}{4\pi} \int_{4\pi} d\Omega P_{LS} = \frac{4}{4\pi} \cdot \frac{G(\Omega') \hat{\omega}_S}{G(\Omega') \hat{\omega}_S} = 1$ .

## §6. Symmetry Relations.

### Diaheliotropic–Paraheliotropic duality.

**Theorem.** For any polar distribution:

$$\Gamma_{LS}^{\text{Dia}} + \Gamma_{LS}^{\text{Para}} = 2 \Gamma_{LS}^{\text{Unif}} \quad (\text{A.19})$$

**Proof.** For Diaheliotropic ( $\eta = 0$ ):  $P_0 = \cos 2\varphi', Q_0 = \sin 2\varphi'$ . For Paraheliotropic ( $\eta = \pi/2$ ):  $P_{\pi/2} = -\cos 2\varphi', Q_{\pi/2} = -\sin 2\varphi'$ . Since the modulation terms enter  $\Gamma_{LS}$  linearly and  $P_0 + P_{\pi/2} = 0, Q_0 + Q_{\pi/2} = 0$ , the sum equals  $2 \times$  (uniform result).

### Planophile–Erectophile complementarity.

**Theorem.** For any azimuthal distribution:

$$\Gamma_{LS}^{\text{Plano}} + 2 \Gamma_{LS}^{\text{Erecto}} = 3 \Gamma_{LS}^{\text{Unif}} \quad (\text{A.20})$$

**Proof.** Planophile:  $(c_0, c_2, c_4) = (3/2, 3/2, 0)$ ; Erectophile:  $(c_0, c_2, c_4) = (3/4, -3/4, 0)$ . Weighted sum:  $c_0$  terms give  $3/2 + 2 \times 3/4 = 3$ ;  $c_2$  terms give  $3/2 + 2 \times (-3/4) = 0$ . Result equals  $3 \times$  (Uniform result).

### Plagiophile–Extremophile complementarity.

**Theorem.** For any azimuthal distribution:

$$\frac{16}{15} \Gamma_{LS}^{\text{Plagi}} + \frac{14}{15} \Gamma_{LS}^{\text{Extrem}} = 2 \Gamma_{LS}^{\text{Unif}} \quad (\text{A.21})$$

**Proof.** Plagiophile:  $(c_0, c_2, c_4) = (15/16, 0, -15/16)$ ; Extremophile:  $(c_0, c_2, c_4) = (15/14, 0, 15/14)$ .  $c_0$  terms:  $(16/15)(15/16) + (14/15)(15/14) = 1+1 = 2$ .  $c_4$  terms:  $(16/15)(-15/16) + (14/15)(15/14) = -1+1 = 0$ .

### §7. Polar Basis Coefficients.

Every polar distribution  $\bar{g}_L(\theta_L)$  is expressible as  $c_0 + c_2 \cos(2\theta_L) + c_4 \cos(4\theta_L)$ . The following Table gives the coefficients.

| Distribution | $\bar{g}_L(\theta_L)$           | $c_0$ | $c_2$ | $c_4$  |
|--------------|---------------------------------|-------|-------|--------|
| Uniform      | 1                               | 1     | 0     | 0      |
| Planophile   | $3\cos^2\theta_L$               | 3/2   | 3/2   | 0      |
| Erectophile  | $\frac{3}{2}\sin^2\theta_L$     | 3/4   | -3/4  | 0      |
| Plagiophile  | $\frac{15}{8}\sin^2(2\theta_L)$ | 15/16 | 0     | -15/16 |
| Extremophile | $\frac{15}{7}\cos^2(2\theta_L)$ | 15/14 | 0     | 15/14  |